F18CF – Linear Algebra

Tutorial Sheet 7: Inner Product Spaces

Getting started.

1. (a) The projection $\vec{v}$ is given by

$$\vec{v} = \frac{\langle \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \rangle}{\langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \rangle} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{6}{3} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}.$$

(b) We have

$$\langle \vec{v}_1, \vec{v}_2 \rangle = \langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \rangle = 1 - 1 + 0 = 0,$$

$$\langle \vec{v}_1, \vec{v}_3 \rangle = \langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \rangle = 1 + 1 - 2 = 0,$$

$$\langle \vec{v}_2, \vec{v}_3 \rangle = \langle \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \rangle = 1 - 1 + 0 = 0.$$

Thus, the family $\vec{v}_1, \vec{v}_2, \vec{v}_3$ is orthogonal. As each vector is non-zero, it follows that it is linearly independent ("an orthogonal family of non-zero vectors is linearly independent"). As we have $3 = \dim \mathbb{R}^3$ vectors, it follows that it is a basis of $\mathbb{R}^3$ ("a linearly independent family of n vectors of $\mathbb{R}^n$ is a basis.")
Thus, the decomposition of $(1, 2, 3)^T$ in this orthogonal basis is:

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \frac{\langle \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \rangle}{\langle \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \rangle} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \frac{\langle \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \rangle}{\langle \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \rangle} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + \frac{\langle \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \rangle}{\langle \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix} \rangle} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix},$$

hence

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}.$$

(c) We have

$$\langle \vec{v}_1, \vec{v}_2 \rangle = \langle \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix} \rangle = 0 + 1 - 1 + 0 = 0,$$

$$\langle \vec{v}_1, \vec{v}_3 \rangle = \langle \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \rangle = 1 + 0 - 1 + 0 = 0,$$

$$\langle \vec{v}_2, \vec{v}_3 \rangle = \langle \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \rangle = 0 + 0 + 1 - 1 = 0.$$

Thus, the family $\vec{v}_1, \vec{v}_2, \vec{v}_3$ is orthogonal.

Thus, the orthogonal projection $\vec{v}$ of $(1,1,1,1)^T$ on $\text{span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$ is given by

$$\vec{v} = \frac{\langle \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix} \rangle}{\langle \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix} \rangle} \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix} + \frac{\langle \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix} \rangle}{\langle \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix} \rangle} \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix}$$

$$+ \frac{\langle \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \rangle}{\langle \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} \rangle} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix},$$

hence

$$\vec{v} = \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ -1 \\ 0 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4/3 \\ 2/3 \\ 1 \\ 2/3 \end{pmatrix}.$$

2. (a) The matrix associated to the system $a_1\vec{v}_1 + a_2\vec{v}_2 + a_3\vec{v}_3 = \vec{0}$ is

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix},$$

which is triangular with non-zero diagonal coefficients, hence has no free variable. Thus the only solution to the system $a_1\vec{v}_1 + a_2\vec{v}_2 + a_3\vec{v}_3 = \vec{0}$ is $a_1 = a_2 = a_3 = 0$, and the family is linearly independent. As this linearly independent family has exactly $3 = \dim \mathbb{R}^3$ vectors, it follows that it is a basis. ("a linearly independent family of n vectors of $\mathbb{R}^n$ is a basis.")

We apply the Gram-Schmidt algorithm to obtain an orthogonal basis $\vec{e}_1, \vec{e}_2, \vec{e}_3$. We have

$$\vec{e}_1 = \vec{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix},$$

$$\vec{e}_2 = \vec{v}_2 - \frac{\langle \vec{v}_2, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle}\vec{e}_1 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{2}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2/3 \\ 1/3 \\ 1/3 \end{pmatrix}.$$

$$\vec{e}_3 = \vec{v}_3 - \frac{\langle \vec{v}_3, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle}\vec{e}_1 - \frac{\langle \vec{v}_3, \vec{e}_2 \rangle}{\langle \vec{e}_2, \vec{e}_2 \rangle}\vec{e}_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{3}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \frac{1/3}{2/3}\begin{pmatrix} -2/3 \\ 1/3 \\ 1/3 \end{pmatrix} = \begin{pmatrix} 0 \\ -1/2 \\ 1/2 \end{pmatrix}.$$

Thus, an orthogonal basis of $\mathbb{R}^3$ containing $\vec{v}_1$ is:

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -2/3 \\ 1/3 \\ 1/3 \end{pmatrix}, \begin{pmatrix} 0 \\ -1/2 \\ 1/2 \end{pmatrix}.$$

(b) The matrix associated to the system $a_1\vec{v}_1 + a_2\vec{v}_2 + a_3\vec{v}_3 + a_4\vec{v}_4 = \vec{0}$ is

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 2 & -1 & 2 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 2 & -1 & 2 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & -1 & 2 & 0 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

$$\rightsquigarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \rightsquigarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 4 \end{pmatrix}.$$

As the row echelon form has no free variable, the only solution is $a_1 = a_2 = a_3 = a_4 = 0$, and the family is linearly independent. As this linearly independent

family has exactly $4 = \dim \mathbb{R}^4$ vectors, it follows that it is a basis. ("a linearly independent family of n vectors of $\mathbb{R}^n$ is a basis.")

We apply the Gram-Schmidt algorithm to obtain an orthogonal basis $\vec{e}_1, \vec{e}_2, \vec{e}_3$. We have

$$\vec{e}_1 = \vec{v}_1 = \begin{pmatrix} 0 \\ 2 \\ 1 \\ 0 \end{pmatrix},$$

$$\vec{e}_2 = \vec{v}_2 - \frac{\langle \vec{v}_2, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle} \vec{e}_1 = \begin{pmatrix} 1 \\ -1/5 \\ 2/5 \\ 0 \end{pmatrix}.$$

$$\vec{e}_3 = \vec{v}_3 - \frac{\langle \vec{v}_3, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle} \vec{e}_1 - \frac{\langle \vec{v}_3, \vec{e}_2 \rangle}{\langle \vec{e}_2, \vec{e}_2 \rangle} \vec{e}_2 = \begin{pmatrix} 1/2 \\ 1/2 \\ -1 \\ -1 \end{pmatrix}.$$

$$\vec{e}_4 = \vec{v}_4 - \frac{\langle \vec{v}_4, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle} \vec{e}_1 - \frac{\langle \vec{v}_4, \vec{e}_2 \rangle}{\langle \vec{e}_2, \vec{e}_2 \rangle} \vec{e}_2 - \frac{\langle \vec{v}_4, \vec{e}_3 \rangle}{\langle \vec{e}_3, \vec{e}_3 \rangle} \vec{e}_3 = \begin{pmatrix} 4/15 \\ 4/15 \\ -8/15 \\ 4/5 \end{pmatrix}.$$

Thus, an orthogonal basis of $\mathbb{R}^4$ containing $\vec{v}_1$ is:

$$\begin{pmatrix} 0 \\ 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1/5 \\ 2/5 \\ 0 \end{pmatrix}, \begin{pmatrix} 1/2 \\ 1/2 \\ -1 \\ -1 \end{pmatrix}, \begin{pmatrix} 4/15 \\ 4/15 \\ -8/15 \\ 4/5 \end{pmatrix}.$$

3. (a) For $\vec{x} = (x_1, x_2, x_3)^T$, $\vec{y} = (y_1, y_2, y_3)^T$, $\vec{z} = (z_1, z_2, z_3)^T$, and $\lambda \in \mathbb{R}$, we have:

- $\langle \vec{x}, \vec{y} \rangle = x_1 y_1 + 2x_2 y_2 + 3x_3 y_3 = y_1 x_1 + 2y_2 x_2 + 3y_3 x_3 = \langle \vec{y}, \vec{x} \rangle$.
- $\langle \vec{x} + \vec{z}, \vec{y} \rangle = (x_1 + z_1) y_1 + 2(x_2 + z_2) y_2 + 3(x_3 + z_3) y_3 = (x_1 y_1 + 2x_2 y_2 + 3x_3 y_3) + (z_1 y_1 + 2z_2 y_2 + 3z_3 y_3) = \langle \vec{x}, \vec{y} \rangle + \langle \vec{z}, \vec{y} \rangle$.
- $\langle \lambda \vec{x}, \vec{y} \rangle = (\lambda x_1) y_1 + (\lambda x_2) y_2 + (\lambda x_3) y_3 = \lambda (x_1 y_1 + 2x_2 y_2 + 3x_3 y_3) = \lambda \langle \vec{x}, \vec{y} \rangle$.
- $\langle \vec{x}, \vec{x} \rangle = x_1^2 + 2x_2^2 + 3x_3^2 \geq 0$. Moreover, a sum of real squares is zero if and only if each term is zero so $\langle \vec{x}, \vec{x} \rangle = 0 \Rightarrow \vec{x} = \vec{0}$.

Thus, $\langle \cdot, \cdot \rangle$ defines an inner product.

(b) For $f, g, h \in \mathcal{C}_{2\pi}(\mathbb{R})$ and $\lambda \in \mathbb{R}$, we have

- $\langle f, g \rangle = \int_0^{2\pi} f(t) g(t) \mathrm{d}t = \int_0^{2\pi} g(t) f(t) \mathrm{d}t = \langle g, f \rangle$.
- $\langle f + g, h \rangle = \int_0^{2\pi} (f(t) + g(t)) h(t) \mathrm{d}t = \int_0^{2\pi} f(t) h(t) \mathrm{d}t + \int_0^{2\pi} g(t) h(t) \mathrm{d}t = \langle f, h \rangle + \langle g, h \rangle$.
- $\langle \lambda f, g \rangle = \int_0^{2\pi} (\lambda f(t)) g(t) \mathrm{d}t = \lambda \int_0^{2\pi} f(t) g(t) \mathrm{d} = \lambda \langle f, g \rangle$.

• $\langle f, f\rangle = \int_0^{2\pi} f^2(t)\mathrm{d}t \geq 0$, as the integral of a non-negative function. Moreover, the integral of a continuous non-negative function on $[0, 2\pi]$ is zero if and only if $f(t) = 0$ for every $t \in [0, 2\pi]$. (cf. Calculus course) Since f is 2π-periodic, this implies that $f(t) = 0$ for every $t \in \mathbb{R}$. Thus, $\langle f, f\rangle = 0 \Rightarrow f$ is the zero function. Thus, $\langle \cdot, \cdot \rangle$ defines an inner product.

Problems.

4. **Computing projections.**

(a) H is a subspace of $\mathbb{R}^4$ since it is the solution space of a homogeneous system of linear equations. The associated matrix is

$$(1\ 1\ 1\ 1),$$

which has three free variables, so we set $y = \alpha, z = \beta, t = \gamma$, and the general solution is thus of the form

$$\begin{pmatrix} -\alpha-\beta-\gamma \\ \alpha \\ \beta \\ \gamma \end{pmatrix} = \alpha \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

Hence a basis of H is

$$\begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

(b) We apply the Gram-Schmidt algorithm to obtain an orthogonal basis $\vec{e}_1, \vec{e}_2, \vec{e}_3$ of H. We have

$$\vec{e}_1 = \vec{v}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix},$$

$$\vec{e}_2 = \vec{v}_2 - \frac{\langle \vec{v}_2, \vec{e}_1\rangle}{\langle \vec{e}_1, \vec{e}_1\rangle}\vec{e}_1 = \begin{pmatrix} -1/2 \\ -1/2 \\ 1 \\ 0 \end{pmatrix}.$$

$$\vec{e}_3 = \vec{v}_3 - \frac{\langle \vec{v}_3, \vec{e}_1\rangle}{\langle \vec{e}_1, \vec{e}_1\rangle}\vec{e}_1 - \frac{\langle \vec{v}_3, \vec{e}_2\rangle}{\langle \vec{e}_2, \vec{e}_2\rangle}\vec{e}_2 = \begin{pmatrix} -1/3 \\ -1/3 \\ -1/3 \\ 1 \end{pmatrix}.$$

(c) (i) The orthogonal projection of $\vec{v}$ on H is given by

$$\pi_H(\vec{v}) = \frac{\langle \vec{v}, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle}\vec{e}_1 + \frac{\langle \vec{v}, \vec{e}_2 \rangle}{\langle \vec{e}_2, \vec{e}_2 \rangle}\vec{e}_2 + \frac{\langle \vec{v}, \vec{e}_3 \rangle}{\langle \vec{e}_3, \vec{e}_3 \rangle}\vec{e}_3 = \begin{pmatrix} -3/2 \\ -1/2 \\ 1/2 \\ 3/2 \end{pmatrix}.$$

(ii) The orthogonal projection of $\vec{v}$ on H is given by

$$\pi_H(\vec{v}) = \frac{\langle \vec{v}, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle}\vec{e}_1 + \frac{\langle \vec{v}, \vec{e}_2 \rangle}{\langle \vec{e}_2, \vec{e}_2 \rangle}\vec{e}_2 + \frac{\langle \vec{v}, \vec{e}_3 \rangle}{\langle \vec{e}_3, \vec{e}_3 \rangle}\vec{e}_3 = \begin{pmatrix} 1/2 \\ -1/2 \\ -1/2 \\ 1/2 \end{pmatrix}.$$

5. **A least squares problem.**

(a) The associated system is

$$\underbrace{\begin{pmatrix} 4 & -2 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{pmatrix}}_{=A} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 7 \\ 0 \\ 0 \end{pmatrix}.$$

This system is inconsistent since $(R_2) - (R_4)$ yields $b = 0$, (R_3) yields $c = 7$, and plugging in these numbers in (R_1) and (R_2) yields $a = -7/4$ and $a = -7$ respectively, which is impossible.

Alternative way: By the Fundamental Theorem of Algebra, a non-zero polynomial of degree ≤ 2 has at most 2 roots. Since $P(-2) = P(-1) = P(1) = P(2) = 0$, P has at least 4 roots, and it follows that P is the zero polynomial. But this is absurd, since $P(0) = 7$. Thus, there is no polynomial of degree ≤ 2 satisfying all our equations.

(b) The least squares solution to this problem is the solution of the equation

$$A^T A\vec{x} = A^T \begin{pmatrix} 0 \\ 0 \\ 7 \\ 0 \\ 0 \end{pmatrix}.$$

We have

$$A^T A = \begin{pmatrix} 34 & 0 & 10 \\ 0 & 10 & 0 \\ 10 & 0 & 5 \end{pmatrix}$$

and

$$A^T \begin{pmatrix} 0 \\ 0 \\ 7 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 7 \end{pmatrix}$$

so the new system to solve is

$$\left(\begin{array}{ccc|c} 34 & 0 & 10 & 0 \\ 0 & 10 & 0 & 0 \\ 10 & 0 & 5 & 7 \end{array}\right) \rightsquigarrow \left(\begin{array}{ccc|c} 34 & 0 & 10 & 0 \\ 0 & 10 & 0 & 0 \\ 0 & 0 & 1 & 3.4 \end{array}\right)$$

By substitution, we get $c = 3.4, b = 0, a = -1$. So the least squares solution of the system

$$\underbrace{\begin{pmatrix} 4 & -2 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{pmatrix}}_{=A} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 7 \\ 0 \\ 0 \end{pmatrix}$$

is

$$\begin{pmatrix} -1 \\ 0 \\ 3.4 \end{pmatrix}.$$

In other words, the best approximating polynomial is $-x^2 + 3.4$

6. **Approximation by polynomials.**

(a) For $f, g, h \in \mathcal{P}_2(\mathbb{R})$ and $\lambda \in \mathbb{R}$, we have

• $\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)\mathrm{d}t = \int_{-1}^{1} g(t)f(t)\mathrm{d}t = \langle g, f\rangle$.

• $\langle f + g, h\rangle = \int_{-1}^{1}(f(t) + g(t))h(t)\mathrm{d}t = \int_{-1}^{1} f(t)h(t)\mathrm{d}t + \int_{-1}^{1} g(t)h(t)\mathrm{d}t = \langle f, h\rangle + \langle g, h\rangle$.

• $\langle \lambda f, g\rangle = \int_{-1}^{1}(\lambda f(t))g(t)\mathrm{d}t = \lambda \int_{-1}^{1} f(t)g(t)\mathrm{d} = \lambda\langle f, g\rangle$.

• $\langle f, f\rangle = \int_{-1}^{1} f^2(t)\mathrm{d}t \geq 0$, as the integral of a non-negative function. Moreover, a polynomial is a continuous function, and we know that the integral of a continuous non-negative function on $[-1, 1]$ is zero if and only if $f(t) = 0$ for every $t \in [-1, -1]$. (cf. Calculus course) But a polynomial vanishes on the interval $[-1, 1]$ if and only if it is the zero polynomial. Thus, $\langle f, f\rangle = 0 \Rightarrow f$ is the zero polynomial. Thus, $\langle \cdot, \cdot \rangle$ defines an inner product.

(b) A basis of of $\mathcal{P}_2(\mathbb{R})$ is $\vec{v}_1 = 1, \vec{v}_2 = x, \vec{v}_3 = x^2$. We apply the Gram-Schmidt algorithm to obtain an orthogonal basis $\vec{e}_1, \vec{e}_2, \vec{e}_3$, recalling that $\int_{-1}^{1} t^k dt = \frac{2}{k+1}$ if k is even and 0 otherwise:

$$\vec{e}_1 = 1,$$

$$\vec{e}_2 = \vec{v}_2 - \frac{\langle \vec{v}_2, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle} \vec{e}_1 = x.$$

$$\vec{e}_3 = \vec{v}_3 - \frac{\langle \vec{v}_3, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle} \vec{e}_1 - \frac{\langle \vec{v}_3, \vec{e}_2 \rangle}{\langle \vec{e}_2, \vec{e}_2 \rangle} \vec{e}_2 = x^2 - 1/3.$$

(c) So the orthogonal projection of x^3 onto $\mathcal{P}_2(\mathbb{R})$ is

$$\frac{\langle x^3, \vec{e}_1 \rangle}{\langle \vec{e}_1, \vec{e}_1 \rangle} \vec{e}_1 + \frac{\langle x^3, \vec{e}_2 \rangle}{\langle \vec{e}_2, \vec{e}_2 \rangle} \vec{e}_2 + \frac{\langle x^3, \vec{e}_3 \rangle}{\langle \vec{e}_3, \vec{e}_3 \rangle} \vec{e}_3 = \frac{3}{5} x.$$

7. **Orthogonal functions.**

(a) We have $\sin(kx) = \frac{e^{ikx} - e^{-ikx}}{2i}$, so

$$\begin{aligned} \sin(kx) \times \sin(lx) &= \frac{e^{ikx} - e^{-ikx}}{2i} \times \frac{e^{ilx} - e^{-ilx}}{2i} = -\frac{1}{4}((e^{i(k+l)x} + e^{-i(k+l)x}) - (e^{i(k-l)x} + e^{-i(k-l)x})) \\ &= \frac{1}{2}(\cos((k-l)x) - \cos((k+l)x)) \end{aligned}$$

(b) For distinct $k, l \geq 1$, we thus have

$$\begin{aligned} \langle \sin(kx), \sin(lx) \rangle &= \int_0^{2\pi} \frac{1}{2}(\cos(\underbrace{(k-l)}_{\neq 0} t) - \cos(\underbrace{(k+l)}_{\neq 0} t))dt \\ &= \frac{1}{2}[\frac{1}{k-l} \sin((k-l)x)]_0^{2\pi} - \frac{1}{2}[\frac{1}{k+l} \sin((k+l)x)]_0^{2\pi} = 0. \end{aligned}$$

Thus, the family $(\sin(nx))_{n \geq 1}$ is orthogonal.

The norm of the function $\sin(nx)$ is

$$\langle \sin(nx), \sin(nx) \rangle^{1/2} = \Big(\int_0^{2\pi} \frac{1}{2}(1 - \cos(\underbrace{(2n)}_{\neq 0} t))dt \Big)^{1/2} = \Big(\pi - \frac{1}{2}[\frac{1}{2n} \sin(2nx)]_0^{2\pi} \Big)^{1/2} = \sqrt{\pi}.$$